

Homework Assignment 1

Empirical Analysis for Policymakers

Sciences Po Paris - Reims

Leave your answers as comments in your code for specific questions. Submit **a single .zip file containing your R script.**

Exercise 1

Download the `boston.csv` data set from Moodle. Import it into R but be careful, the first row contains data. Make sure to **untick** "First Row as Names" while importing the data. Then, run the code below to assign column names for the variables:

```
colnames(boston) <- c('CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM', 'AGE', 'DIS', 'RAD', 'TAX',  
'PTRATIO', 'B', 'LSTAT', 'MEDV')
```

Each record in the database describes a Boston suburb or town. Take a look at the [metadata](#) to get an understanding of the data and variables.

Basic R Syntax

1. Load the dataset `boston.csv` and display the first 10 rows.
2. Summarize the data using `summary()`. What is the mean per capita crime rate by town (**CRIM**)? What is the maximum ratio of pupils to teachers (**PTRATIO**) across all towns?
3. The government wants to provide financial assistance to households in ten suburbs that are **not** near the Charles River (**CHAS** = 0). They will conduct a lottery to select these suburbs. Write a function called `select_suburbs()` that takes the dataset as input, filters the suburbs that are not by the Charles River, and randomly selects ten suburbs (hint: try using `sample_n` instead of `sample` for data tables).

dplyr for Data Manipulation

4. Using `dplyr`, filter the data to show towns where the proportion of residential land zoned for lots over 25,000 sq. ft. (**ZN**) is greater than 20 and the average number of rooms per dwelling (**RM**) is

above 6. Display the first 5 rows of the filtered data.

5. Create a new column that shows the tax rate (**TAX**) per 10,000 dollars and filter the dataset to show rows where this value is less than 20. How many towns meet this criterion?
6. Use `mutate()` to create a new column representing the log transformation of the median value of owner-occupied homes (**MEDV**). What is the average log-transformed **MEDV** for homes that have a proportion of owner-occupied units built before 1940 (**AGE**) above 80?
7. Group the data by the number of accessibility to radial highways (**RAD**) and calculate the average crime rate (**CRIM**) for each group. Which group has the lowest average crime rate?

ggplot2 for Data Visualization

8. Create a scatter plot using `ggplot2` to visualize the relationship between the number of rooms per dwelling (**RM**) and the median value of homes (**MEDV**). Add a regression line to the plot and describe the relationship observed.
9. Plot a histogram of the per capita crime rate (**CRIM**). Adjust the bin width to 0.2, set limits for the x-axis between 0 and 20 and describe the distribution.
10. Create a boxplot comparing the distribution of the median value of homes (**MEDV**) for towns that border the Charles River (**CHAS** = 1) versus those that do not (**CHAS** = 0). What does this plot suggest about the impact of proximity to the river on home values?
11. Create a density plot of the proportion of lower status of the population (**LSTAT**). Highlight the area under the curve where **LSTAT** is above 20. What does the plot indicate about the distribution of this variable?